

BOSTON HOUSE PRICE PREDICTION

Using Multiple Linear Regression

Machine Learning Course

MIT Applied AI and Data Science Program

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Executive Summary

This project developed a multiple linear regression model to predict housing prices in Boston neighborhoods using a set of demographic, environmental, accessibility, and socioeconomic variables. Exploratory data analysis identified several variables with skewed distributions and strong intercorrelations, motivating the application of a logarithmic transformation to the target variable and the assessment of multicollinearity among predictors.

The final regression model retained eight statistically significant predictors of housing prices: CRIM, CHAS, NOX, RM, DIS, RAD, PTRATIO, and LSTAT. The model explained approximately 76% of the variability in housing prices, indicating that these factors play an important role in determining residential property values. Further analyses demonstrated that key assumptions of linear regression were reasonably satisfied, while model evaluation on the test dataset yielded a Mean Absolute Percentage Error (MAPE) of approximately 5.3%, indicating strong predictive performance.

The results highlight the importance of socioeconomic conditions, housing characteristics, environmental quality and accessibility in shaping residential property prices. These insights can support data-driven decision-making in real estate investment, urban planning and development, and housing market analysis.

Business Problem Overview

Estimating housing prices is a critical problem for a wide range of stakeholders, including homeowners, real estate professionals, investors, lending institutions, urban planners, policymakers and government agencies. Housing values are influenced by a variety of factors like neighborhood characteristics, environmental conditions, accessibility to infrastructure and employment centers, and socioeconomic conditions. Understanding how these factors affect housing values can support more informed decisions on property investments, mortgage lending, urban development, and housing market assessments.

The objective of this analysis is to identify the key factors that influence housing prices in Boston neighborhoods and to develop a predictive model capable of estimating median home values. By quantifying the relationships between housing prices and neighborhood characteristics, the analysis provides insights that can help stakeholders better evaluate properties, assess market conditions, and prioritize investment and development opportunities.

To address this problem, a multiple linear regression approach was employed. The analysis began with exploratory data analysis (EDA) to understand the distribution and relationships among variables, followed by data preprocessing, including a logarithmic transformation of the target variable and multicollinearity assessment using Variance Inflation Factors (VIF). The regression model was subsequently refined through feature selection and evaluated using statistical diagnostics and performance metrics. This methodology provides a framework for estimating housing values based on neighborhood characteristics while identifying the factors most strongly associated with residential property prices.

Data Overview

The data used in this analysis is the Boston Housing dataset, which consists of 506 observations and 13 variables including 12 predictor variables and one target variable. The target variable, MEDV, represents the median value of owner-occupied homes and serves as the response variable in the analysis. The predictor variables include measures such as crime rate, residential zoning, industrial land use, air pollution, average number of rooms, housing age, accessibility to employment centers and highways, property tax rate, pupil-teacher ratio, and socioeconomic status (Table 1).

All variables in the dataset are numeric. The variable **CHAS** is a binary indicator that identifies whether a neighborhood borders the Charles River.

Table 1. Dataset used in this analysis.

Variable	Description
CRIM	Per capita crime rate by town
ZN	Proportion of residential land zoned for large lots
INDUS	Proportion of non-retail business acres per town
CHAS	Charles River dummy variable
NOX	Nitric oxide concentration
RM	Average number of rooms per dwelling
AGE	Proportion of homes built before 1940
DIS	Weighted distance to employment centers
RAD	Accessibility to radial highways
TAX	Property tax rate
PTRATIO	Pupil-teacher ratio
LSTAT	Percentage of lower-status population
MEDV	Median value of owner-occupied homes (target variable)

EDA and Data Preprocessing

- **Data Quality Assessment**

The dataset was initially examined to identify potential data quality issues that could affect the analysis. No missing values or duplicate records were detected, indicating that the dataset was complete and suitable for analysis without requiring imputation, record removal, or other corrective procedures.

- **Summary Statistics and Univariate Analysis**

Summary statistics (Appendix 1), and histograms were used to evaluate the distribution and variability of the variables. The target variable, **MEDV**, exhibits an approximately bell-shaped distribution, although it is not perfectly normal. Most housing values are concentrated between 15 and 30, while the noticeable spike at 50 suggests a possible upper-value cap in the dataset (Figure 1).

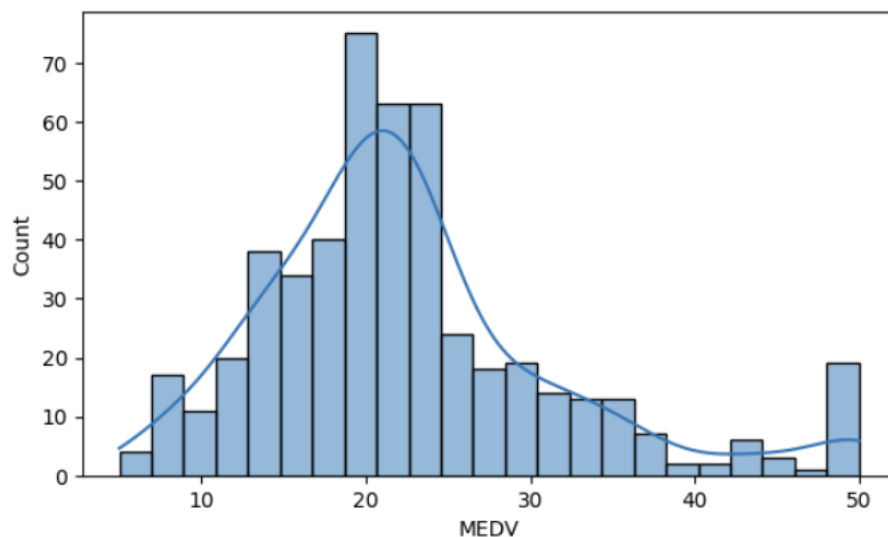


Figure 1. Distribution of MEDV variable.

Several predictor variables, including **CRIM**, **ZN**, **DIS**, and **LSTAT**, are positively skewed and contain potential extreme observations. In contrast, **INDUS**, **RAD**, and **TAX** exhibit bimodal distributions, suggesting the presence of distinct neighborhood groups.

Variables such as **RM** (Figure 2) and **PTRATIO** appear relatively symmetric, whereas **CHAS** is a binary variable. Overall, the distributions indicate that the dataset contains a mixture of approximately normal, skewed, bimodal, and binary variables, reflecting the diverse characteristics of the neighborhoods represented in the data.

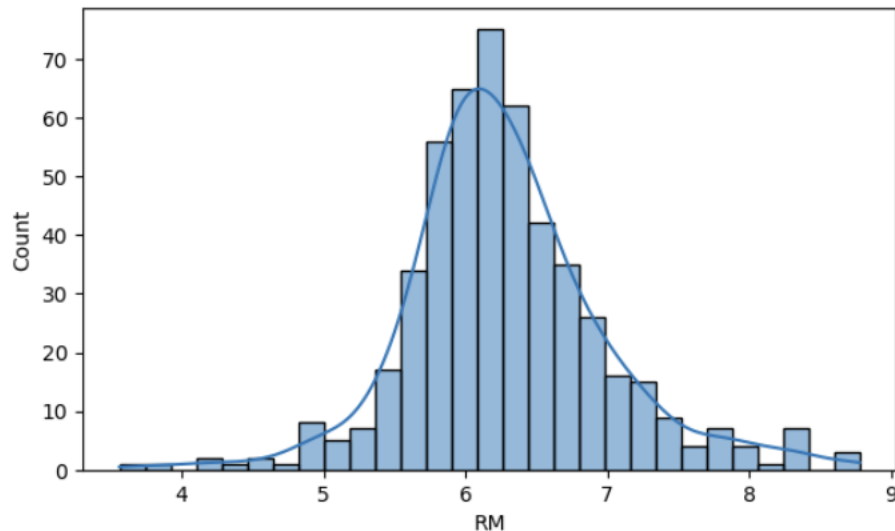


Figure 2. Distribution of RM variable.

The summary statistics further revealed substantial variability among neighborhoods. Variables such as **ZN**, **RAD**, **TAX**, and **LSTAT** exhibited wide ranges and relatively large standard deviations, indicating considerable differences in zoning practices, accessibility, taxation levels, and socioeconomic conditions. The median value of **AGE** (77.5) suggests that many neighborhoods contain a high proportion of homes built before 1940, while the large differences between the median and maximum values of **CRIM** and **LSTAT** indicate the presence of skewness and potential extreme observations.

- **Outlier Assessment**

The univariate analysis identified potential outliers in several variables, particularly **CRIM**, **ZN**, **DIS**, and **LSTAT**, as evidenced by their long-tailed distributions and large differences between median and maximum values. Additional investigation during the bivariate analysis revealed that the strong correlation between **RAD** and **TAX** was partially influenced by observations associated with very high tax rates. When observations with $TAX \geq 600$ were temporarily filtered for exploratory purposes, the

correlation between these variables decreased substantially, suggesting that extreme tax values contributed to the observed relationship.

Because these observations likely represented valid neighborhood characteristics rather than data entry errors, they were retained in the dataset. Instead of removing these observations, their impact was addressed during model development through multicollinearity assessment and feature selection.

- **Data Transformation**

Because the target variable exhibited slight positive skewness, a logarithmic transformation was applied to MEDV. The transformed variable (MEDV_log) displayed a more symmetric and approximately normal distribution, reducing skewness and improving its suitability for linear regression modeling (Figure 3).

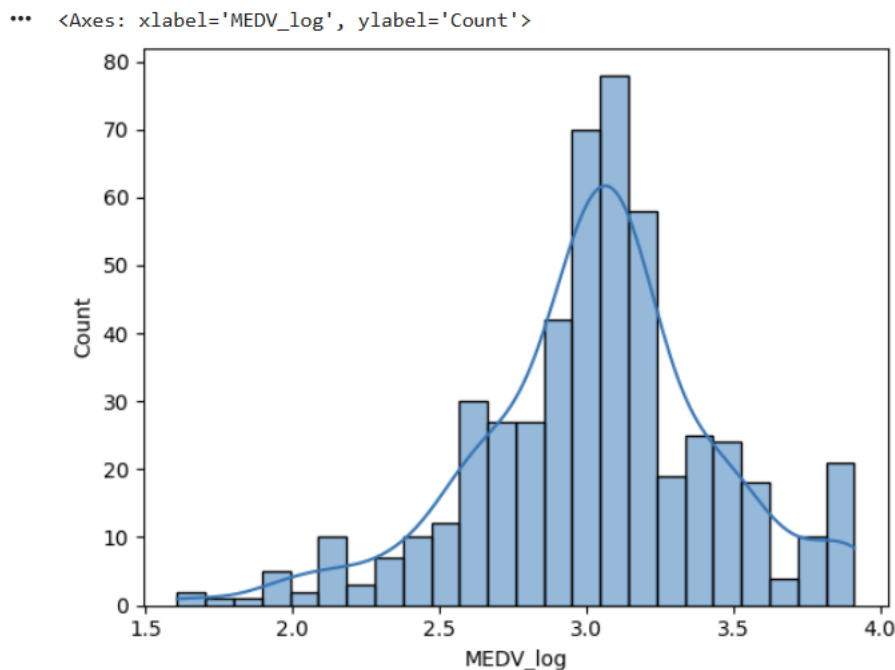


Figure 3. Distribution of the log-transformed variable MEDV_log.

- **Bivariate Analysis**

A correlation heatmap (Fig. 4) and scatterplots were used to evaluate relationships among variables and identify potential multicollinearity issues. The

analysis revealed several meaningful relationships between the predictor variables and the target variable (**MEDV_log**). Among all predictors, **LSTAT** exhibited the strongest negative correlation with housing prices (-0.81), indicating that neighborhoods with a higher proportion of lower-status population tend to have lower housing values. In contrast, **RM** showed the strongest positive correlation (0.65), suggesting that neighborhoods with larger homes are generally associated with higher housing values.

Additional variables, including **CRIM**, **INDUS**, **NOX**, **TAX**, and **PTRATIO**, displayed moderate negative correlations with housing values, indicating that higher crime rates, pollution levels, tax rates, and socioeconomic disadvantage are generally associated with lower property values.

The heatmap also revealed strong relationships among several predictor variables. In particular, **RAD** and **TAX** exhibited a very strong positive correlation (0.91), suggesting potential multicollinearity. Strong associations were also observed among **NOX**, **INDUS**, **AGE**, and **DIS**, highlighting the interconnected nature of environmental, land-use, and accessibility characteristics.

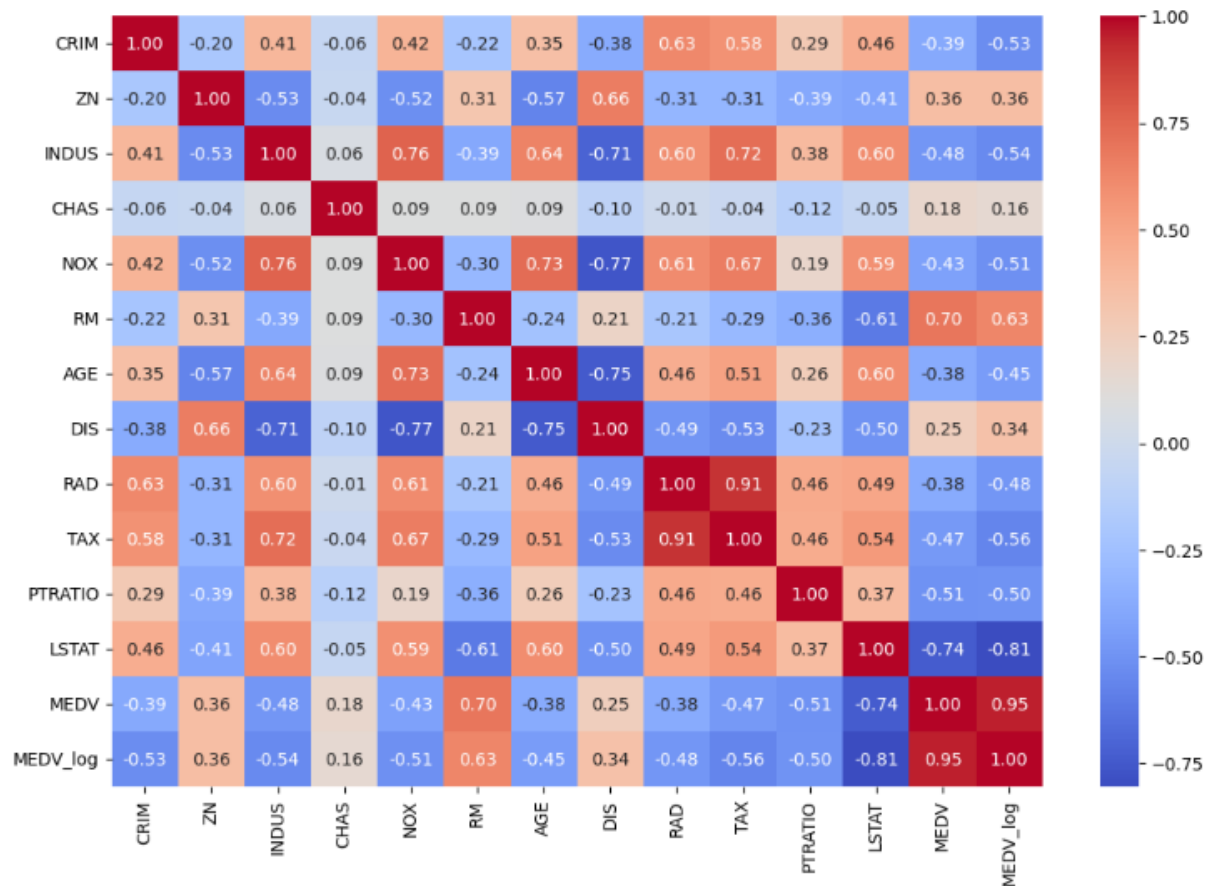


Figure 4. Correlation heatmap.

Scatterplots further confirmed these relationships. The relationship between **RAD** and **TAX** showed a strong positive correlation. However, visual inspection suggested that this strong correlation may have been influenced by extreme observations associated with high tax rates. After filtering observations with **TAX** ≥ 600 , the correlation between **RAD** and **TAX** decreased substantially, indicating that the original high correlation was partly driven by these high-tax observations. Since these values likely represent valid neighborhood-level characteristics rather than data entry errors, they were not removed from the dataset. Also, **INDUS** and **NOX** exhibited a strong positive association, indicating that neighborhoods with greater industrial land use tend to have higher levels of air pollution (Fig. 5)

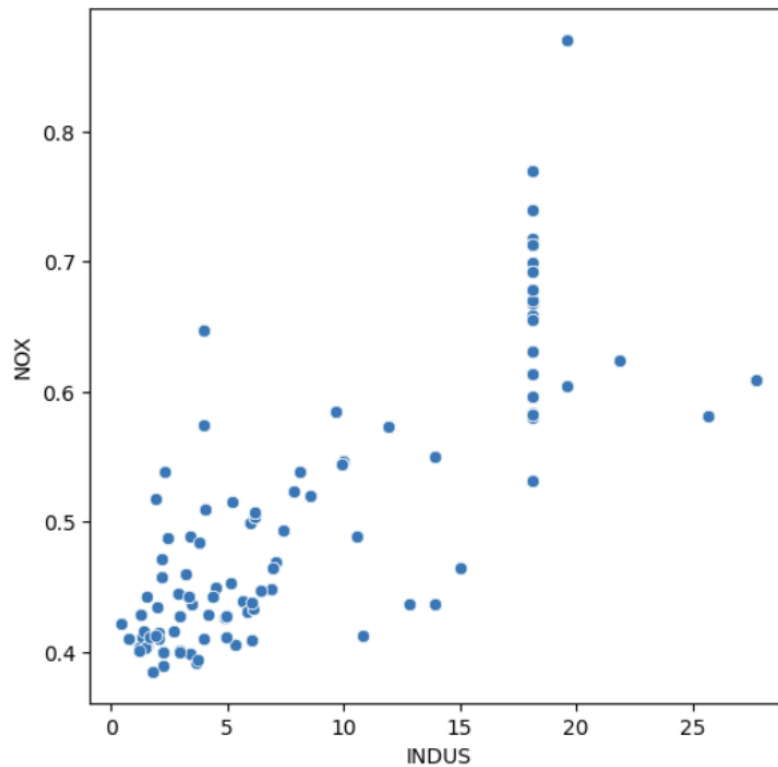


Figure 5. INDUS and NOX scatterplot.

Similarly, **AGE** and **NOX** were positively related, suggesting that neighborhoods with a greater proportion of homes built before 1940 tend to have a higher NOX concentration. This pattern is consistent with the strong positive correlation observed in the heatmap ($r = 0.73$) and may reflect the tendency of older urbanized areas to experience higher levels of air pollution (Fig. 6).

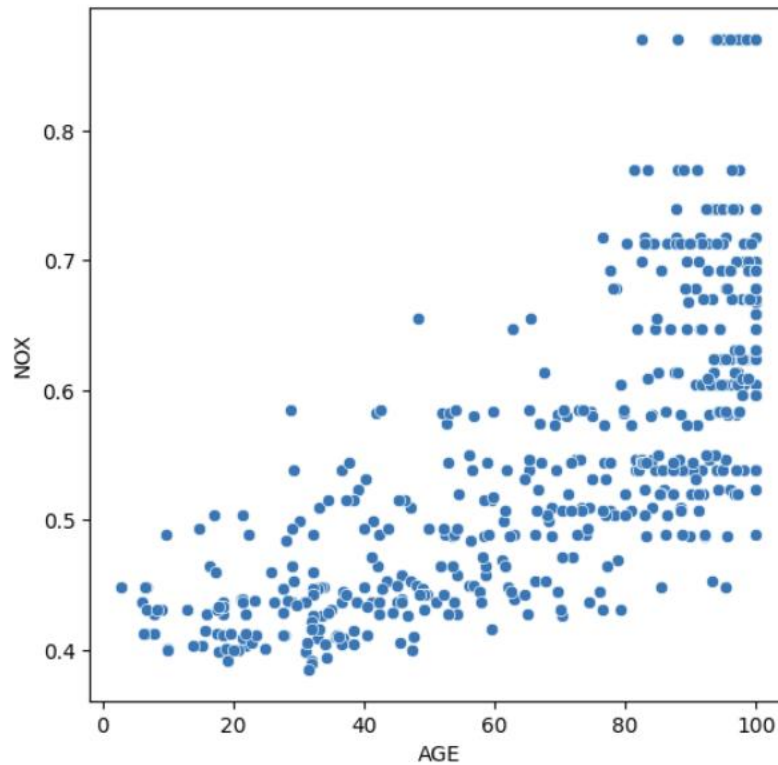


Figure 6. AGE versus NOX scatterplot.

In contrast, **DIS** and **NOX (Fig. 7)** displayed a strong negative relationship ($r = -0.77$), indicating that neighborhoods located farther from employment centers tend to experience lower pollution levels. This pattern suggests that areas closer to urban and industrial centers may be more exposed to traffic and industrial emissions, resulting in higher nitric oxide concentrations.

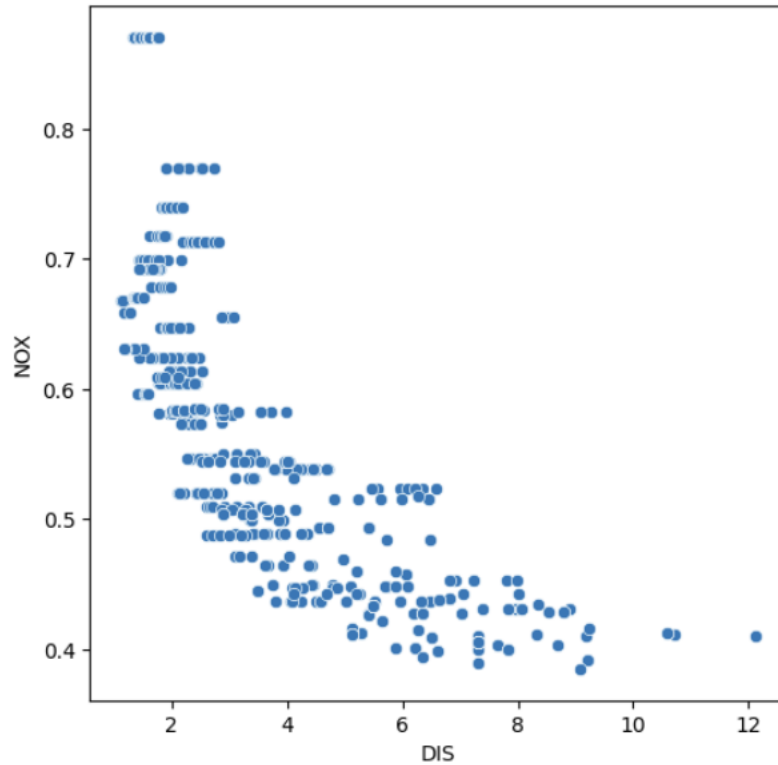


Fig 7. NOX versus DIS scatterplot.

Overall, the correlation and scatterplot analyses identified several highly related predictors, suggesting the presence of multicollinearity and motivating a formal assessment using Variance Inflation Factors (VIF) prior to model development.

- **Multicollinearity Assessment and Feature Selection**

To formally assess multicollinearity, Variance Inflation Factors (VIF) were calculated. At first, the variable TAX exhibited the highest VIF value and was removed from the model to reduce redundancy among predictors. After recalculating the VIF values, all remaining variables exhibited acceptable levels of multicollinearity.

A subsequent OLS regression analysis identified ZN, AGE, and INDUS as statistically insignificant predictors with a p -values greater than 0.05. These variables were removed to obtain a more parsimonious and interpretable model. The final set of predictors retained for modeling consisted of CRIM, CHAS, NOX, RM, DIS, RAD, PTRATIO, and LSTAT.

Linear Regression Analysis and its Assumptions

A multiple linear regression model was developed using the **Ordinary Least Squares (OLS)** estimation method to quantify the relationship between housing prices and neighborhood characteristics. OLS estimates the regression coefficients by minimizing the sum of squared residuals, thereby identifying the linear combination of predictors that best explains variation in the target variable.

As mentioned before, the target variable (**MEDV**) was log-transformed to improve normality and better satisfy the assumptions of linear regression. The dataset was then divided into training and testing subsets using a 70:30 split. To ensure the reliability of the coefficient estimates, multicollinearity among predictors was assessed using Variance Inflation Factors (VIF), while model assumptions were evaluated through residual diagnostics and statistical tests.

After evaluating multicollinearity and statistical significance, variables exhibiting high redundancy or limited explanatory contribution were removed. Specifically, **TAX** was excluded due to its high VIF value, while **ZN**, **AGE**, and **INDUS** were removed because they were not statistically significant predictors. The final model retained eight predictors: **CRIM**, **CHAS**, **NOX**, **RM**, **DIS**, **RAD**, **PTRATIO**, and **LSTAT**.

The final OLS model (Appendix 3), achieved an R^2 of 0.767 and an Adjusted R^2 of 0.762, indicating that approximately 76% of the variability in housing prices can be explained by the selected predictors. See coefficient table 2.

Table 2. Coefficients table.

Variable	Coefficient
Intercept	4.6494
CRIM	-0.0125
CHAS	0.1198
NOX	-1.0562
RM	0.0589

DIS	-0.0441
RAD	0.0078
PTRATIO	-0.0485
LSTAT	-0.0293

The coefficient estimates indicate that **RM** and **CHAS** are positively associated with housing values, whereas **CRIM**, **NOX**, **DIS**, **PTRATIO**, and **LSTAT** exhibit negative relationships with housing prices. Among the retained variables, **LSTAT** showed the strongest negative association with housing values, while **RM** exhibited the strongest positive association. These findings suggest that socioeconomic conditions, housing characteristics, environmental quality, and accessibility play important roles in determining residential property values.

- **Testing the Assumptions of the Linear Regression Model**

Because Ordinary Least Squares regression relies on several statistical assumptions, diagnostic tests were conducted to evaluate the validity of the final model.

- **Multicollinearity Assessment**

Multicollinearity was initially assessed using a correlation heatmap and subsequently quantified using Variance Inflation Factors (VIF). The variable **TAX** exhibited the highest VIF value and was removed from the model. After recalculating the VIF scores (Appendix 2), all remaining predictors displayed acceptable levels of multicollinearity, indicating that the predictor variables provided distinct information and did not excessively inflate coefficient estimates.

- **Residual Mean**

The mean of the residuals was calculated to verify that prediction errors were centered around zero. The residual mean was approximately -2.53×10^{-15} , which is approximately zero. This indicates that the model does not systematically overestimate or underestimate housing values.

- **Homoscedasticity**

The Goldfeld–Quandt test was used to evaluate whether the residual variance remained constant across observations (Table 3). In this context, homoscedasticity means that the model errors are spread out fairly evenly across the range of predicted housing values.

Table 3. Goldfeld–Quandt Test Results

Statistic	Value
F-statistic	1.0835
p-value	0.3019

The Goldfeld–Quandt test produced a p-value of **0.3019**, which is greater than the significance level of **0.05**. Therefore, the test did not find evidence of heteroscedasticity. This indicates that the residuals appear to have approximately constant variance across the range of fitted values. As a result, the homoscedasticity assumption of the linear regression model is reasonably satisfied.

- **Linearity**

Linearity was assessed using residual-versus-fitted plot (Fig. 8) and LOWESS smoothing curves. The residuals appeared randomly scattered around zero with no strong systematic patterns or pronounced curvature. The LOWESS curve remained relatively flat across the range of fitted values, suggesting that a linear relationship adequately describes the relationship between the predictors and housing prices.

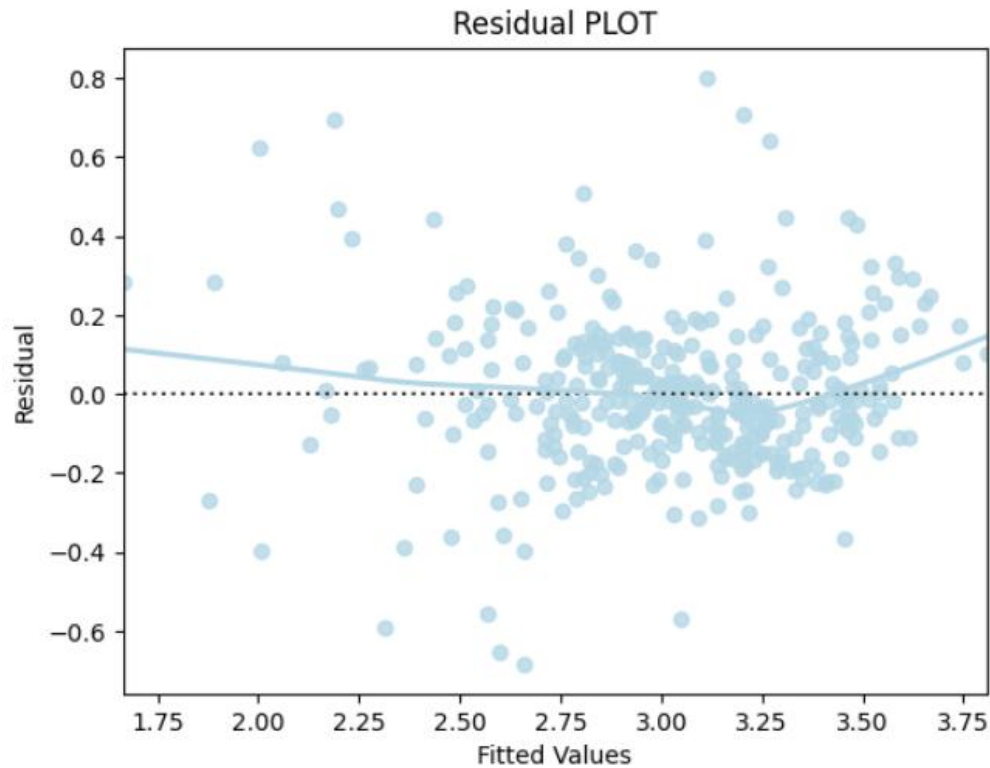


Figure 8. Residual plot.

- **Normality of Residuals**

The normality assumption was evaluated using residual histograms and Q-Q plots (Fig. 9). The residual distribution appeared approximately bell-shaped and centered around zero. The Q-Q plot showed that most observations closely followed the reference line, although minor deviations were observed in the tails. Overall, the residuals were considered sufficiently close to a normal distribution for the purposes of linear regression analysis.

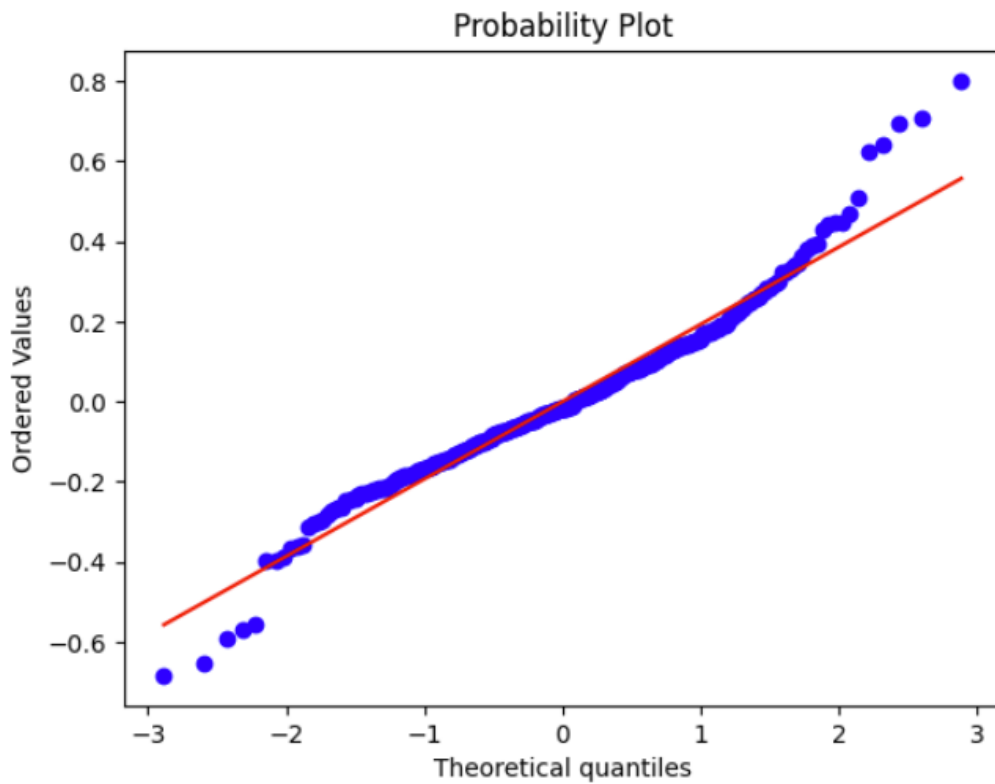


Figure 9. Q-Q plot.

The diagnostic tests indicate that the major assumptions of linear regression were reasonably satisfied, supporting the reliability of the coefficient estimates and the validity of the model's statistical inferences.

- **Model Performance Evaluation**

The predictive performance of the final regression model was evaluated using multiple metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). These metrics were calculated for both the training and testing datasets to assess the model's ability to generalize to unseen observations (Table 4).

Table 4. Model Performance on Training and Testing Data

Dataset	RMSE	MAE	MAPE (%)
Training	0.1955	0.1437	4.98
Testing	0.1980	0.1513	5.26

The model achieved low prediction errors on both the training and testing datasets. The similarity between the training and testing metrics indicates that the model generalizes well to unseen data and does not exhibit substantial overfitting. In particular, the test MAPE of approximately **5.26%** suggests that the model's predictions differ from the actual housing values by only a small percentage on average.

- **Cross-Validation**

To further evaluate the robustness and stability of the model, ten-fold cross-validation was performed. Cross-validation reduces dependence on a single train-test split by repeatedly fitting and evaluating the model on different subsets of the data (Table 5).

Table 5. Cross-Validation Results

Metric	Mean	Standard Deviation
R^2	0.729	0.232
Mean Squared Error (MSE)	0.041	0.023

The cross-validation results indicate that the model maintains consistent predictive performance across different subsets of the dataset. The average **R^2 of 0.729** demonstrates that the model explains a substantial proportion of the variability in housing prices even when evaluated on multiple validation folds. Furthermore, the relatively low average MSE indicates that prediction errors remain small across different partitions of the data.

Overall, the model demonstrated strong predictive performance, good generalization capability, and stable results under cross-validation. These findings suggest that the final regression model is both reliable and suitable for estimating housing values based on neighborhood characteristics.

Conclusions

The multiple linear regression analysis successfully identified the neighborhood characteristics most strongly associated with housing values in Boston. The final model explained approximately **76% of the variation in housing prices** (Adjusted $R^2 = 0.762$) and demonstrated strong predictive performance on unseen data, indicating that the selected variables capture many of the key factors influencing residential property values.

The regression coefficients provide valuable insights into the drivers of housing prices. Among the retained predictors, **LSTAT** exhibited the strongest negative association with housing values, indicating that neighborhoods with higher levels of socioeconomic disadvantage tend to have lower property values. Conversely, **RM** showed the strongest positive association, suggesting that larger homes are generally associated with higher housing values. Environmental quality (**NOX**), public safety (**CRIM**), educational conditions (**PTRATIO**), and accessibility-related characteristics (**DIS**, **RAD**, and **CHAS**) were also found to significantly influence housing prices.

Business Recommendations

Based on the findings of the regression analysis, the following actions are recommended:

- **Consider socioeconomic conditions when evaluating investment opportunities.** The strong negative relationship between **LSTAT** and housing values suggests that neighborhoods with lower socioeconomic disadvantage tend to exhibit higher property values. While this information can support investment decisions, it should be balanced with broader community development goals and opportunities for growth in historically underserved areas.

- **Focus on properties with larger living spaces.** The positive association between **RM** and housing values indicates that homes with a greater number of rooms are generally associated with higher market values and stronger returns.
- **Consider environmental quality when evaluating investment opportunities.** The negative relationship between **NOX** and housing values suggests that neighborhoods with lower pollution levels may be more attractive to buyers and support higher property prices.
- **Incorporate crime and safety indicators into valuation decisions.** Since **CRIM** was negatively associated with housing values, investments in safer neighborhoods may reduce risk and improve long-term appreciation potential.
- **Support improvements in educational resources.** The negative association between **PTRATIO** and housing values indicates that neighborhoods with stronger educational conditions tend to be more desirable and may sustain higher property values.
- **Leverage accessibility and location advantages.** Variables such as **DIS**, **RAD**, and **CHAS** demonstrate that transportation access, proximity to employment centers, and desirable geographic features can significantly influence housing values. These factors should be considered when evaluating development and investment opportunities.

Key Business Takeaways

- Housing values are strongly influenced by a combination of socioeconomic, environmental, housing, and accessibility factors.
- Socioeconomic conditions (**LSTAT**) and housing size (**RM**) emerged as the most influential predictors of property values.
- The final regression model explained approximately **76% of the variation in housing prices**, providing a reliable framework for property valuation and market analysis.

- The insights generated by the model can support more informed investment decisions, improve property valuation accuracy, reduce uncertainty, and help stakeholders allocate resources more effectively.

The findings of this analysis can support a variety of stakeholders in the housing market. Real estate investors and developers can use the identified predictors to evaluate investment opportunities, while financial institutions can incorporate these factors into property valuation and risk assessment processes. Similarly, urban planners and policymakers can leverage the results to guide investments in environmental quality, public safety, educational resources, and transportation infrastructure.

Overall, the analysis demonstrates that housing values are influenced by a combination of socioeconomic, environmental, housing, and accessibility factors. By identifying and quantifying these relationships, the regression model provides a practical framework for improving property valuation, reducing investment uncertainty, and supporting data-driven decision-making in real estate and urban development. Although the model explains a substantial proportion of the variability in housing values, approximately 24% of the variation remains unexplained, suggesting that additional factors not included in the dataset may also influence residential property values.

Appendix

Appendix 1. Summary statistics

	count	mean	std	min	25%	50%	75%	max
CRIM	506.0	3.613524	8.601545	0.006320	0.082045	0.256510	3.677083	88.976200
ZN	506.0	11.363636	23.322453	0.000000	0.000000	0.000000	12.500000	100.000000
INDUS	506.0	11.136779	6.860353	0.460000	5.190000	9.690000	18.100000	27.740000
CHAS	506.0	0.069170	0.253994	0.000000	0.000000	0.000000	0.000000	1.000000
NOX	506.0	0.554695	0.115878	0.385000	0.449000	0.538000	0.624000	0.871000
RM	506.0	6.284634	0.702617	3.561000	5.885500	6.208500	6.623500	8.780000
AGE	506.0	68.574901	28.148861	2.900000	45.025000	77.500000	94.075000	100.000000
DIS	506.0	3.795043	2.105710	1.129600	2.100175	3.207450	5.188425	12.126500
RAD	506.0	9.549407	8.707259	1.000000	4.000000	5.000000	24.000000	24.000000
TAX	506.0	408.237154	168.537116	187.000000	279.000000	330.000000	666.000000	711.000000
PTRATIO	506.0	18.455534	2.164946	12.600000	17.400000	19.050000	20.200000	22.000000
LSTAT	506.0	12.653063	7.141062	1.730000	6.950000	11.360000	16.955000	37.970000
MEDV	506.0	22.532806	9.197104	5.000000	17.025000	21.200000	25.000000	50.000000
MEDV_log	506.0	3.034513	0.408757	1.609438	2.834680	3.054001	3.218876	3.912023

Appendix 2. VIF results

```
# Check for VIF  
print(checking_vif(X_train))
```

...		feature	VIF
	0	const	532.025529
	1	CRIM	1.923159
	2	ZN	2.483399
	3	INDUS	3.270983
	4	CHAS	1.050708
	5	NOX	4.361847
	6	RM	1.857918
	7	AGE	3.149005
	8	DIS	4.333734
	9	RAD	2.942862
	10	PTRATIO	1.909750
	11	LSTAT	2.860251

Appendix 3. OLS model results.

OLS Regression Results						
Dep. Variable:	MEDV_log	R-squared:	0.767			
Model:	OLS	Adj. R-squared:	0.762			
Method:	Least Squares	F-statistic:	142.1			
Date:	Fri, 05 Jun 2026	Prob (F-statistic):	2.61e-104			
Time:	19:01:06	Log-Likelihood:	75.486			
No. Observations:	354	AIC:	-133.0			
Df Residuals:	345	BIC:	-98.15			
Df Model:	8					
Covariance Type: nonrobust						
	coef	std err	t	P> t	[0.025	0.975]
const	4.6494	0.242	19.242	0.000	4.174	5.125
CRIM	-0.0125	0.002	-7.349	0.000	-0.016	-0.009
CHAS	0.1198	0.039	3.093	0.002	0.044	0.196
NOX	-1.0562	0.168	-6.296	0.000	-1.386	-0.726
RM	0.0589	0.020	2.928	0.004	0.019	0.098
DIS	-0.0441	0.008	-5.561	0.000	-0.060	-0.028
RAD	0.0078	0.002	3.890	0.000	0.004	0.012
PTRATIO	-0.0485	0.006	-7.832	0.000	-0.061	-0.036
LSTAT	-0.0293	0.002	-12.949	0.000	-0.034	-0.025
Omnibus:	32.514	Durbin-Watson:	1.925			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	87.354			
Skew:	0.408	Prob(JB):	1.07e-19			
Kurtosis:	5.293	Cond. No.	690.			